

Technical Datasheet - VBF-T5R1-DSH-01

VBF-T5R1 delivery package - generated 2026-08-25

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Product: NMC811/graphite(-SiOx) high-energy pouch cell, design R7B (case t5_r1).

Generation date: 2026-08-25. Values mechanically from parameter sets / simulation outputs / log-evaluate entries; dual-system (Chen2020 / OKane2022) columns given where they differ.

Field	Value	Source
Rated (nominal) capacity	5.0 Ah	`Nominal cell capacity [A.h]` (both parameter sets)
Verified 1C discharge capacity	6.210 Ah (Chen) / 6.209 Ah (OKane), DFN	`cell/r7b_*_1c_dfn.json`; design exceeds nominal (N/P 1.40, anode-unlocked)
Nominal voltage	3.798 V (Chen) / 3.591 V (OKane) discharge midpoint	`calc-energy` midpoint_voltage_v
Voltage window	2.5 ? 4.2 V	parameter sets
Rated energy	22.236 Wh (Chen) / 22.202 Wh (OKane)	`calc-energy` energy_wh (1C time-integrated V*I)
Gravimetric energy density	642.1 Wh/kg (Chen) / 641.1 Wh/kg (OKane) - contract caliber, electrolyte excluded	`calc-energy` energy_density_wh_kg
Volumetric energy density	1064.5 Wh/L (Chen) / 1062.9 Wh/L (OKane) - contract caliber, electrolyte excluded	`calc-energy` energy_density_wh_l
Mass (contract caliber, electrolyte excluded)	34.631 g	`calc-energy` mass_kg
Electrolyte mass (reference)	9.53 g (pore volume ? 1.2 g/cm3 literature density)	mechanical, design_spec ?3
Maximum continuous discharge	1C (= 5 A nominal current), verified 6.21 Ah to 2.5 V	1C_discharge protocol, 25 ?C
Fast-charge capability	4C (20 A) CC from 2.5 V to 4.2 V at 45 ?C ambient: T_max 326.8 K (53.7 ?C, Chen) / 328.2 K (55.0 ?C, OKane) <= 60 ?C; no lithium plating (anode surface potential min +0.0183 V / +0.0180 V)	`4C_charge_45C` protocol, lumped thermal + plating, DFN
4C charge acceptance	4.73 Ah (Chen) / 5.00 Ah (OKane) physical	charge-segment duration ? 20 A ? 3600 (mechanical; library capacity_ah key under-reports 5?, see caveat)
Operating temperature range	Simulated at 25 ?C (1C discharge) and 45 ?C (4C charge). Wider range not simulated	protocol definitions; honest statement
Cycle life	Not simulated in this case (aging protocol outside task contract). Chen2020/OKane2022 are aging-capable parameter sets; a 100-cycle aging run was not executed - value not fabricated.	honest annotation

Field	Value	Source
Safety determination	4C fast charge: no plating, temperature below 60 °C red line - PASS under both parameter systems. Abuse scenarios (nail, overcharge-to-TR, crush) not simulated	dvpr.md
Dimensions	65 mm × 1580 mm × 203.4 μm layer stack; shell thickness Not provided	parameter sets + calc-energy
DC resistance (calc-energy caliber)	3.50 mΩ (Chen) / 17.93 mΩ (OKane)	`calc-energy` dcr_ohm

Caveats (honest disclosure):

1. Energy/mass numbers exclude electrolyte (contract caliber, `electrolyte\_included: false`).
2. The 4C-output `capacity\_ah` key is under-reported ~5% by a pybamm_runner library bug; physical charge acceptance is computed separately and stated above.
3. Electrolyte transport values (≈ 2.0 S/m, $D_e \approx 4.5 \times 10^{-10} \text{ m}^2/\text{s}$, $t \approx 0.45$) are design targets at literature-upper-bound estimates - formulation must be validated experimentally.
4. Plating margin is +18 mV - thin by simulation standards; manufacturing tolerances must preserve the 2.61 μm negative particle size and the stated porosity/thickness.
5. No true DFT/MD endorsement (real_compute=false, recorded in log endorse entry).